

BUFFALO NIAGARA, NY, USA

WMO: 725280

Lat: 42.9411N

Lon: 78.7189W

Elev: 715

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.5	6.8	-5.3	4.2	3.8	-1.2	5.3	8.2	33.1	31.6	29.9	32.0	10.4	260	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	86.3	71.3	83.9	70.0	81.7	69.1	74.7	81.7	73.2	80.3	71.6	78.5	11.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	122.9	78.9	70.7	116.0	77.6	69.2	110.1	76.3	38.6	81.7	37.2	80.6	35.9	78.6	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	23.8	20.5	-2.4	90.6	5.4	2.7	-6.3	92.5	-9.4	94.0	-12.5	95.5	-16.4	97.5
			-3.1	76.5	5.1	1.9	-6.8	77.9	-9.8	79.0	-12.7	80.1	-16.4	81.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	25.2	26.5	34.0	45.6	57.8	67.0	71.6	70.3	63.8	52.1	41.1	31.5
	DBStd	18.56	11.35	10.52	10.48	8.97	8.30	6.50	5.07	4.94	6.96	8.17	8.81	9.26
	HDD50	3092	771	661	507	189	21	0	0	0	1	73	292	577
	HDD65	6449	1233	1079	960	583	255	53	7	11	104	407	717	1040
	CDD50	2722	2	1	12	57	262	511	668	631	414	138	24	2
	CDD65	606	0	0	0	1	31	113	210	177	67	7	0	0
	CDH74	4110	0	0	4	25	238	769	1470	1118	450	36	0	0
	CDH80	865	0	0	0	4	42	173	330	221	93	2	0	0
Wind	WSAvg	10.0	12.0	11.6	10.8	10.7	9.7	9.0	8.7	8.0	8.2	9.5	10.5	11.0
Precipitation	PrecAvg	38.7	3.0	2.3	2.8	3.0	3.3	3.3	3.1	3.8	3.6	3.1	3.8	3.6
	PrecMax	53.5	6.9	5.9	6.0	5.9	7.2	8.3	8.9	10.7	9.0	6.1	9.8	8.7
	PrecMin	28.5	1.1	0.8	1.2	0.9	0.6	0.6	0.7	1.7	0.8	0.3	1.5	1.5
	PrecStd	5.2	1.4	1.1	1.1	1.3	1.4	1.8	1.7	1.8	1.7	1.3	1.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	59.8	69.9	77.5	84.3	88.8	89.3	88.7	87.1	78.1	67.6	60.7
		MCWB	55.0	51.9	57.2	61.6	68.0	72.0	72.4	71.5	71.2	65.6	57.4	54.7
	2%	DB	52.0	51.8	60.9	71.0	80.2	84.5	86.3	84.9	82.6	73.7	63.1	52.9
		MCWB	48.7	46.0	52.1	57.6	66.4	69.7	71.9	71.0	69.7	63.3	55.6	48.2
	5%	DB	45.5	45.2	55.3	65.9	76.2	81.5	83.6	82.3	79.2	69.4	59.3	48.8
		MCWB	42.0	40.3	48.4	54.1	63.1	68.1	70.3	69.6	67.8	61.3	51.8	45.3
	10%	DB	40.3	40.0	49.8	60.8	72.1	78.6	81.3	80.0	75.8	65.3	54.8	44.1
		MCWB	37.2	36.4	43.5	51.3	60.8	66.9	69.6	68.8	66.1	58.0	49.4	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	53.4	59.1	64.1	70.5	75.4	76.9	76.0	74.2	69.1	59.7	54.9
		MCDB	60.4	58.4	68.2	74.7	81.5	82.9	84.4	82.7	82.3	74.3	65.1	60.1
	2%	WB	49.3	46.3	53.4	59.4	68.1	72.7	75.1	74.2	71.9	65.6	56.7	49.7
		MCDB	52.1	51.2	58.9	68.2	77.7	81.0	82.1	80.6	79.4	72.0	61.7	52.1
	5%	WB	42.7	41.0	48.9	56.1	65.3	70.7	73.4	72.5	69.7	61.9	53.3	45.5
		MCDB	45.2	44.6	55.2	63.8	74.1	78.3	80.1	79.0	76.4	68.0	58.2	49.1
	10%	WB	37.4	36.4	43.9	52.2	62.6	69.0	71.9	71.0	67.8	59.1	49.9	40.8
		MCDB	39.9	39.8	49.1	59.1	69.7	75.9	78.5	77.3	74.2	64.6	54.6	43.7
Mean Daily Temperature Range	5% DB	MDBR	12.5	13.1	14.5	17.4	17.7	16.3	16.0	16.0	16.6	14.6	12.7	10.8
		MCDBR	17.8	20.7	23.0	26.1	22.0	20.0	18.1	18.6	20.0	19.8	19.3	16.4
		MCWBR	15.9	16.3	15.7	16.2	11.6	9.8	8.7	9.1	10.1	11.5	14.2	14.1
	5% WB	MCDBR	17.5	19.9	21.7	23.5	20.1	16.3	14.6	14.4	16.5	16.9	17.6	16.3
		MCWBR	16.2	16.8	15.9	16.7	12.0	9.6	8.5	9.0	9.5	11.4	14.3	14.8
Clear-Sky Solar Irradiance	taub	0.295	0.321	0.331	0.371	0.400	0.433	0.432	0.422	0.375	0.358	0.312	0.295	
	taud	2.396	2.319	2.406	2.334	2.291	2.230	2.245	2.261	2.374	2.414	2.511	2.466	
	Ebn at Noon	267	277	289	284	277	267	266	265	270	260	260	258	
	Edh at Noon	25	31	33	38	41	44	43	41	34	29	22	21	
All-Sky Solar Radiation	RadAvg	410	663	1038	1395	1712	1838	1856	1651	1315	794	492	333	
	RadStd	39	66	110	149	179	142	145	90	119	75	69	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page